

# Regional Robustness Across Vision Representation Paradigms on GeoDE and Dollar Street

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## Abstract

Modern vision models achieve strong benchmark performance, but average accuracy can hide uneven behavior across geographic regions. This paper studies that tension using nine object-recognition models grouped into four representation paradigms: Convolutional Vision, Transformer Vision, Self-supervised Foundation, and Vision-Language Foundation. The evaluation uses GeoDE and Dollar Street, two datasets designed to expose geographically diverse visual contexts. For each model and dataset, the paper compares ImageNet-pretrained evaluation against evaluation after fine-tuning on the dataset training split. The analysis reports overall accuracy, per-region accuracy, worst-region accuracy, best-region accuracy, and a geographic-disparity measure defined as the best-region accuracy minus the worst-region accuracy. In the current results, fine-tuning improves average object accuracy for all 18 dataset-model pairs, but it also increases measured geographic disparity for all 18 pairs. ConvNeXtV2 achieves the highest fine-tuned accuracy on both datasets, while MAE has the lowest fine-tuned disparity on both datasets. These findings suggest that high recognition accuracy and regional robustness are related but distinct evaluation targets. The results are descriptive and do not establish causal mechanisms.

## 1 Introduction

Object recognition has improved rapidly as visual representation learning has moved from classical convolutional networks to large-scale transformer, self-supervised, and vision-language systems. The practical question, however, is not only whether a model is accurate on average. A model deployed in geographically diverse settings may encounter familiar object labels in unfamiliar forms: different materials, household layouts, camera conditions, cultural practices, and socioeconomic contexts. In that setting, average accuracy can hide regional failure modes. A model can look strong overall while performing substantially worse for one region than another.

This paper studies that gap between benchmark progress and regional robustness. The central question is whether modern representation paradigms reduce geographic disparity, reproduce it, or reveal different versions of it when evaluated on globally diverse household-object datasets. The paper therefore treats model architecture and pretraining style as a lens for analysis rather than as a leaderboard. The evaluated models are grouped into four paradigms: the Convolutional Vision Paradigm, represented by ResNet50 and ConvNeXtV2; the Transformer Vision Paradigm, represented by DeiT3, SwinV2, and MaxViT; the Self-supervised Foundation Paradigm, represented by DINOv2 and MAE; and the Vision-Language Foundation Paradigm, represented by CLIP and SigLIP.

GeoDE and Dollar Street are appropriate datasets for this question because both were built to make geographic visual variation visible. GeoDE is a geographically diverse object-recognition

dataset with images collected across six world regions [Ramaswamy et al., 2023]. Dollar Street contains household-object images from homes around the world, along with demographic metadata such as region, country, and monthly income [Gaviria Rojas et al., 2022]. Prior Dollar Street work also motivates the concern that object-recognition systems may not perform uniformly across geographic and socioeconomic contexts [DeVries et al., 2019]. These datasets are therefore useful not merely as alternative test sets, but as probes of whether object-recognition performance is distributed evenly across regions.

Average accuracy is insufficient for this paper’s question because it collapses regional behavior into a single number. If one model performs well in Europe but worse in Africa, and another model has lower average accuracy but smaller regional variation, the models pose different evaluation trade-offs. This paper therefore reports object accuracy together with per-region accuracy, worst-region accuracy, best-region accuracy, and geographic disparity. The goal is not to label one model family as fair or unfair, but to test whether the same model that improves average recognition also improves regional balance.

Comparing the four representation paradigms is meaningful because they encode different assumptions and training signals. Convolutional models emphasize locality and hierarchical feature extraction. Transformer models use attention mechanisms that can aggregate information differently across spatial positions. Self-supervised foundation models learn visual representations without ordinary supervised class labels. Vision-language foundation models align images with natural language, potentially drawing on broader semantic supervision. These differences may be associated with different regional robustness behavior, although this paper does not claim that a paradigm causes a particular disparity pattern.

The paper makes four contributions. First, it provides a controlled comparison of nine modern vision models across four representation paradigms. Second, it evaluates each model on GeoDE and Dollar Street before and after fine-tuning. Third, it analyzes average accuracy, per-region accuracy, worst-region accuracy, best-region accuracy, geographic disparity, and fine-tuning-induced changes. Fourth, it discusses how regional robustness differs from average object recognition performance and why model families should be compared using both accuracy and disparity.

## 2 Related Work

**Visual representation paradigms and modern recognition backbones.** The models in this paper represent four broad approaches to visual representation learning. ResNet50 is the classical residual convolutional baseline, using skip connections to train deep convolutional networks effectively [He et al., 2016]. ConvNeXtV2 modernizes the convolutional family through architectural and representation-learning changes, including a fully convolutional masked-autoencoder design and Global Response Normalization [Woo et al., 2023]. DeiT3 revisits supervised Vision Transformer training and provides a strong ViT recipe [Touvron et al., 2022]. SwinV2 develops a scalable hierarchical transformer with shifted-window attention and training stabilizations for large-scale vision models [Liu et al., 2022]. MaxViT combines convolutional structure with local-global multi-axis attention [Tu et al., 2022]; this paper places it under the Transformer Vision Paradigm because its central contribution is attention-based aggregation. DINOv2 and MAE provide self-supervised representation-learning baselines, with DINOv2 emphasizing large-scale self-supervised visual features [Oquab et al., 2024] and MAE learning through masked patch reconstruction [He et al., 2022]. CLIP and SigLIP represent vision-language foundation models: CLIP learns transferable visual models through contrastive image-text pretraining [Radford et al., 2021], while SigLIP modifies image-text learning with a sigmoid loss [Zhai et al., 2023]. This paper uses these cita-

tions to describe model families, not to claim that any cited architecture paper establishes regional robustness.

**Dataset bias, domain shift, and geographic robustness.** Dataset bias refers here to systematic skew in collection, representation, labels, or visual context. Early work on dataset bias showed that recognition datasets can contain dataset-specific cues that affect cross-dataset generalization [Torralba and Efros, 2011]. ImageNet and its successors have also motivated questions about whether conventional benchmark progress transfers to distribution-shifted settings [Deng et al., 2009, Recht et al., 2019, Taori et al., 2020]. Domain-shift benchmarks and evaluation suites such as WILDS and DomainBed emphasize that models can behave differently when training and evaluation distributions differ [Koh et al., 2021, Gulrajani and Lopez-Paz, 2021]. Geographic robustness is related but more specific: it asks whether performance remains consistent across regional groups. GeoDE directly frames object recognition as a geographically diverse evaluation problem [Ramaswamy et al., 2023], while Dollar Street provides household-object images with geographic and socioeconomic metadata [Gaviria Rojas et al., 2022]. Prior geodiversity and dataset-audit work supports the premise that geographic representation matters in visual datasets [Shankar et al., 2017, Wang et al., 2022, Kalluri et al., 2023]. These literatures motivate the evaluation design, but they do not by themselves explain the mechanisms behind the empirical gaps observed in this paper.

**Metrics for regional disparity and robustness evaluation.** Robustness evaluation requires more than average accuracy. Worst-group and hidden-stratification work argues that aggregate performance can obscure weak performance for particular groups or subclasses [Sagawa et al., 2020, Sohoni et al., 2020]. Fairness-oriented vision benchmarks similarly motivate reporting performance across groups rather than treating a single aggregate as sufficient [Gustafson et al., 2023]. This paper adapts that intuition to geographic regions by reporting overall accuracy, per-region accuracy, worst-region accuracy, best-region accuracy, and the best-minus-worst geographic-disparity gap. It also reports pretrained-versus-fine-tuned deltas in both accuracy and disparity. The metric is intentionally simple: it identifies uneven regional performance, but it does not prove discrimination, causal mechanisms, or deployment harm. TODO: add confidence intervals and per-region sample counts before the final version.

### 3 Problem Definition and Methodology

Let  $D = \{(x_i, y_i, r_i)\}_{i=1}^n$  be an object-recognition dataset, where  $x_i$  is an image,  $y_i$  is the object label, and  $r_i$  is the geographic region associated with the image. Given a model  $f$ , this paper evaluates whether object-recognition accuracy is consistent across regions. The unit of analysis is a dataset-model-training-state tuple: one dataset, one model, and either the pretrained or fine-tuned state.

Overall object accuracy is defined as

$$\text{Acc}(f) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{f(x_i) = y_i\}.$$

For a region  $r$ , per-region accuracy is

$$\text{Acc}_r(f) = \frac{\sum_{i=1}^n \mathbf{1}\{r_i = r\} \mathbf{1}\{f(x_i) = y_i\}}{\sum_{i=1}^n \mathbf{1}\{r_i = r\}}.$$

This allows the same model to be evaluated both as an average recognizer and as a region-conditioned recognizer.

The paper’s geographic-disparity metric is the maximum difference between regional accuracies:

$$\Delta_{\text{geo}}(f) = \max_r \text{Acc}_r(f) - \min_r \text{Acc}_r(f).$$

This metric follows the same intuition as worst-group evaluation: average accuracy can hide the weakest group performance [Sagawa et al., 2020]. It also follows the regional-evaluation motivation of prior Dollar Street and GeoDE work [DeVries et al., 2019, Ramaswamy et al., 2023]. The formula here is this paper’s adapted best-minus-worst regional gap; it should not be read as a full fairness metric.

The paper compares two training states. The pretrained state evaluates ImageNet-compatible pretrained models before dataset-specific fine-tuning. The fine-tuned state evaluates the same model family after fine-tuning on the training split for each dataset. The fine-tuning change in average accuracy is

$$\Delta_{\text{FT},\text{Acc}} = \text{Acc}(f_{\text{fine-tuned}}) - \text{Acc}(f_{\text{pretrained}}).$$

The fine-tuning change in geographic disparity is

$$\Delta_{\text{FT},\text{Disparity}} = \Delta_{\text{geo}}(f_{\text{fine-tuned}}) - \Delta_{\text{geo}}(f_{\text{pretrained}}).$$

A positive  $\Delta_{\text{FT},\text{Acc}}$  means that fine-tuning improves average accuracy. A negative  $\Delta_{\text{FT},\text{Disparity}}$  means that fine-tuning reduces measured regional disparity. A positive  $\Delta_{\text{FT},\text{Disparity}}$  means that fine-tuning increases measured disparity even if average accuracy improves.

The analysis is descriptive. It can show that high average accuracy and low regional disparity do not always coincide. It can also show that fine-tuned and pretrained states differ. It cannot prove why the difference occurs without additional evidence about class balance, region composition, pretraining data, model capacity, and optimization.

## 4 Datasets

**GeoDE.** GeoDE is a geographically diverse object-recognition dataset containing 61,940 images from 40 classes and six world regions [Ramaswamy et al., 2023]. The paper uses a processed train/validation/test split prepared for this experiment. The local class mapping used by the result files contains 40 classes, matching the official class count. The region labels appearing in the processed results are Africa, Americas, EastAsia, Europe, SouthEastAsia, and WestAsia. GeoDE is useful in this paper because it directly evaluates whether object recognition remains stable across broad geographic regions.

**Dollar Street.** Dollar Street is a supervised household-object dataset with 38,479 images, 289 household-item categories, 404 homes, 63 countries, and four regions [Gaviria Rojas et al., 2022]. The official dataset includes metadata such as region, country, and monthly income. This paper does not use the full 289-category official label space. Instead, it uses a processed ImageNet-compatible experimental subset with 64 classes, as recorded in the local class mapping. The processed result files use four region groups: Africa, Americas, Asia, and Europe. Dollar Street is useful here because everyday household objects can vary strongly in visual form across geography and socioeconomic setting.

Table 1: Dataset summary. Official scale is drawn from the dataset papers; experimental subset details come from the local class mappings and configs.

| Dataset       | Source                  | Official scale                                                        | Experimental subset                   | Classes used | Region groups                                               | Split                 | Purpose in this paper                                                            |
|---------------|-------------------------|-----------------------------------------------------------------------|---------------------------------------|--------------|-------------------------------------------------------------|-----------------------|----------------------------------------------------------------------------------|
| GeoDE         | GeoDE                   | 61,940 images; 40 classes; 6 world regions                            | Processed train/validation/test split | 40           | Africa, Americas, EastAsia, Europe, SouthEastAsia, WestAsia | train/validation/test | Geographically diverse object recognition benchmark                              |
| Dollar Street | MLCommons Dollar Street | 38,479 images; 289 household-item categories; 63 countries; 4 regions | ImageNet-compatible processed subset  | 64           | Africa, Americas, Asia, Europe                              | train/validation/test | Household-object recognition under geographic and socioeconomic visual diversity |

The two datasets play complementary roles. GeoDE is designed specifically around geographically diverse object recognition, while Dollar Street emphasizes household objects embedded in geographic and socioeconomic context. The distinction matters for interpretation: a model that looks regionally stable on one dataset may still show larger geographic disparity on the other. TODO: add local split sizes, per-region sample counts, and class-by-region counts before the final version.

## 5 Models

The paper evaluates nine models grouped into four representation paradigms. The grouping is analytical rather than causal: it helps compare architecture families and pretraining signals, but the results alone do not prove that a paradigm causes a particular regional-disparity pattern.

### 5.1 Convolutional Vision Paradigm

The Convolutional Vision Paradigm contains ResNet50 and ConvNeXtV2. ResNet50 is the classical residual CNN baseline, representing the long-standing convolutional approach to image recognition [He et al., 2016]. ConvNeXtV2 is a modernized ConvNet that combines convolutional architecture design with representation-learning advances, including masked-autoencoder-inspired training ideas [Woo et al., 2023]. This group tests whether convolutional locality and hierarchical feature extraction remain competitive under geographic shift.

### 5.2 Transformer Vision Paradigm

The Transformer Vision Paradigm contains DeiT3, SwinV2, and MaxViT. DeiT3 is used as a strong supervised Vision Transformer recipe [Touvron et al., 2022]. SwinV2 is a hierarchical shifted-window transformer designed for scaling capacity and resolution [Liu et al., 2022]. MaxViT is a hybrid convolution-attention architecture, but this paper places it under the Transformer Vision Paradigm because its central contribution is multi-axis attention [Tu et al., 2022]. This group tests whether attention-based architectures reduce regional sensitivity or primarily improve average recognition.

### 5.3 Self-supervised Foundation Paradigm

The Self-supervised Foundation Paradigm contains DINOv2 and MAE. DINOv2 learns visual features through large-scale self-supervised representation learning [Oquab et al., 2024]. MAE learns by reconstructing masked image patches [He et al., 2022]. These models are useful for the paper because they test whether label-free representation learning transfers more evenly across geographic regions than ordinary supervised recognition pipelines.

### 5.4 Vision-Language Foundation Paradigm

The Vision-Language Foundation Paradigm contains CLIP and SigLIP. CLIP aligns images and text through contrastive language-image pretraining [Radford et al., 2021]. SigLIP uses a sigmoid loss for language-image pretraining [Zhai et al., 2023]. These models test whether semantic supervision from image-text data improves regional consistency, or whether large-scale language-image pretraining still leaves uneven regional behavior.

The taxonomy table describes the intended comparison. The empirical claims in later sections come only from the derived result tables, not from expectations about model families. TODO:

Table 2: Model taxonomy used in the paper.

| Model      | Paradigm                            | Backbone type                             | Pretraining signal                                                      | Expected strength                                      | Possible regional-bias risk                                            |
|------------|-------------------------------------|-------------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------|------------------------------------------------------------------------|
| ResNet50   | Convolutional Vision Paradigm       | Residual CNN                              | ImageNet-style supervised pretraining                                   | Stable classical baseline                              | Local texture/background cues may not transfer evenly across regions   |
| ConvNeXtV2 | Convolutional Vision Paradigm       | Modern ConvNet                            | PCMAE and ImageNet-style fine-tuning                                    | High recognition accuracy with convolutional hierarchy | High average accuracy may still leave regional gaps                    |
| DeiT3      | Transformer Vision Paradigm         | Vision Transformer                        | Supervised ViT training recipe                                          | Strong global representation baseline                  | Attention can improve accuracy without guaranteeing regional balance   |
| SwinV2     | Transformer Vision Paradigm         | Hierarchical shifted-window transformer   | Large-scale supervised/self-supervised windowed-transformer pretraining | Hierarchical attention with local windows              | Windowed attention may still reflect dataset-specific visual content   |
| MaxViT     | Transformer Vision Paradigm         | Hybrid convolution-attention architecture | ImageNet-style supervised pretraining                                   | Local-global multi-axis attention                      | Hybrid design may improve accuracy without eliminating disparity       |
| DINOv2     | Self-supervised Foundation Paradigm | ViT foundation model                      | Self-supervised representation learning                                 | Label-free visual features at scale                    | Pretraining coverage may not match regional household-object variation |
| MAE        | Self-supervised Foundation Paradigm | Masked autoencoding ViT                   | Masked patch reconstruction                                             | Low measured regional disparity in these runs          | Lower disparity can coincide with lower average accuracy               |
| CLIP       | Vision-Language Foundation Paradigm | Image-language ViT                        | Contrastive image-text pretraining                                      | Semantic supervision beyond fixed labels               | Web image-text data can carry uneven geographic coverage               |
| SigLIP     | Vision-Language Foundation Paradigm | Image-language ViT                        | Sigmoid image-text objective                                            | Competitive semantic transfer                          | Language-image supervision does not guarantee regional consistency     |

reconcile the stored Dollar Street SwinV2 fine-tuning configuration before final submission; the result row is labeled with the paper’s intended SwinV2 taxonomy, but one local `used_config.yaml` entry requires audit.

## 6 Experiments

The experiments compare ImageNet-pretrained evaluation with dataset-specific fine-tuning. In the pretrained state, each model is evaluated on the processed GeoDE or Dollar Street test split before dataset-specific training. In the fine-tuned state, the model is initialized from pretrained weights, fine-tuned on the dataset training split, and evaluated on the held-out test split. Results are reported both overall and by region.

The processed datasets use train, validation, and test directories. The local class mappings contain 40 classes for GeoDE and 64 classes for the processed Dollar Street subset. The Dollar Street subset is therefore not the full official 289-category Dollar Street label space; it is an ImageNet-compatible experimental subset. The pretrained evaluation files record that, when the number of target classes differs from 1000, the evaluation uses pretrained weights where shapes match plus a newly initialized dataset classifier head. This detail is important when interpreting the very low pretrained accuracies.

The fine-tuning runs use a common protocol recorded in the local configuration files: input image size 224, batch size 32, Adam optimization, learning rate  $3 \times 10^{-5}$ , cosine scheduling, label smoothing of 0.1, automatic mixed precision, and 10 epochs. Training uses the dataset training split, model selection uses validation performance, and final metrics are computed on the test split. The evaluation script computes object accuracy, region-conditioned object accuracy, best region, worst region, and the best-minus-worst geographic-disparity gap.

The model initialization follows the intended taxonomy in Table 2. The external training script maps the experimental model keys to pretrained model identifiers for ResNet50, ConvNeXtV2, DeiT3, SwinV2, MaxViT, DINOv2, MAE, CLIP, and SigLIP. TODO: before final submission, audit the Dollar Street SwinV2 configuration mismatch noted in the Models section and verify that all result rows correspond to the intended pretrained model identifiers.

All metrics used in the paper are normalized into `derived/processed/all_metrics.csv`. The table contains 36 rows: two datasets, nine models, and two training states. The validation scripts check that all expected rows are present, that fine-tuning state values are valid, and that geographic disparity equals best-region accuracy minus worst-region accuracy. No model training is performed inside the paper workspace.

## 7 Results and Analysis

**Overall accuracy across representation paradigms.** Table 3 shows that fine-tuning improves object accuracy for all 18 dataset-model pairs. On Dollar Street, the largest accuracy increase is for ConvNeXtV2, from 1.6% pretrained accuracy to 76.7% fine-tuned accuracy. On GeoDE, the largest increase is for DeiT3, from 1.8% to 96.3%. The strongest fine-tuned accuracy overall

Table 3: Main comparison of pretrained and fine-tuned accuracy and geographic disparity. Values are percentages.

| Model      | Paradigm                            | Dataset       | Pretrained Acc | Fine-tuned Acc | Acc Change | Pretrained Disparity | Fine-tuned Disparity | Disparity Change |
|------------|-------------------------------------|---------------|----------------|----------------|------------|----------------------|----------------------|------------------|
| ResNet50   | Convolutional Vision Paradigm       | Dollar Street | 2.3            | 58.9           | 56.6       | 0.6                  | 12.2                 | 11.6             |
| ConvNeXtV2 | Convolutional Vision Paradigm       | Dollar Street | 1.6            | 76.7           | 75.1       | 1.3                  | 8.5                  | 7.2              |
| DeiT3      | Transformer Vision Paradigm         | Dollar Street | 1.9            | 75.1           | 73.2       | 0.6                  | 8.9                  | 8.3              |
| SwinV2     | Transformer Vision Paradigm         | Dollar Street | 1.9            | 76.1           | 74.2       | 0.6                  | 7.3                  | 6.6              |
| MaxViT     | Transformer Vision Paradigm         | Dollar Street | 2.9            | 72.9           | 70.0       | 2.5                  | 11.0                 | 8.5              |
| DINOv2     | Self-supervised Foundation Paradigm | Dollar Street | 1.1            | 73.4           | 72.3       | 1.1                  | 7.7                  | 6.6              |
| MAE        | Self-supervised Foundation Paradigm | Dollar Street | 2.6            | 66.7           | 64.1       | 0.7                  | 6.9                  | 6.1              |
| CLIP       | Vision-Language Foundation Paradigm | Dollar Street | 1.5            | 73.1           | 71.6       | 0.6                  | 10.1                 | 9.5              |
| SigLIP     | Vision-Language Foundation Paradigm | Dollar Street | 1.6            | 76.2           | 74.6       | 0.5                  | 7.4                  | 6.9              |
| ResNet50   | Convolutional Vision Paradigm       | GeoDE         | 2.8            | 90.9           | 88.0       | 1.4                  | 2.7                  | 1.3              |
| ConvNeXtV2 | Convolutional Vision Paradigm       | GeoDE         | 2.4            | 96.6           | 94.2       | 1.1                  | 2.3                  | 1.2              |
| DeiT3      | Transformer Vision Paradigm         | GeoDE         | 1.8            | 96.3           | 94.5       | 0.4                  | 2.0                  | 1.6              |
| SwinV2     | Transformer Vision Paradigm         | GeoDE         | 2.0            | 96.3           | 94.2       | 0.7                  | 2.1                  | 1.4              |
| MaxViT     | Transformer Vision Paradigm         | GeoDE         | 2.5            | 95.6           | 93.1       | 1.2                  | 1.9                  | 0.7              |
| DINOv2     | Self-supervised Foundation Paradigm | GeoDE         | 2.1            | 96.0           | 93.9       | 1.1                  | 2.4                  | 1.3              |
| MAE        | Self-supervised Foundation Paradigm | GeoDE         | 3.3            | 92.9           | 89.6       | 1.8                  | 1.9                  | 0.1              |
| CLIP       | Vision-Language Foundation Paradigm | GeoDE         | 4.5            | 95.4           | 90.9       | 1.6                  | 3.1                  | 1.4              |
| SigLIP     | Vision-Language Foundation Paradigm | GeoDE         | 2.5            | 95.9           | 93.4       | 1.1                  | 2.8                  | 1.7              |

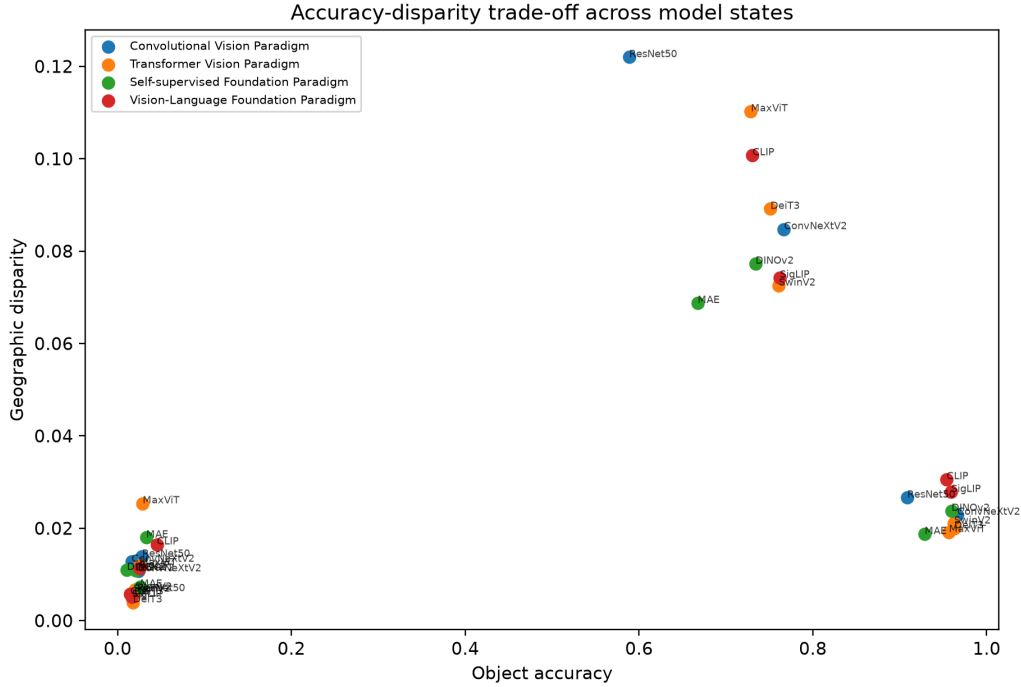


Figure 1: Accuracy-disparity trade-off across model states. Each point is a dataset-model-state evaluation. The plot shows why high average accuracy and low geographic disparity should be evaluated jointly.

is ConvNeXtV2 on GeoDE at 96.6%, and ConvNeXtV2 is also the highest-accuracy fine-tuned model on Dollar Street at 76.7%. At the paradigm level, the Transformer Vision Paradigm has the highest mean fine-tuned accuracy across the two datasets at 85.4%, closely followed by the Vision-Language Foundation Paradigm at 85.1%. This suggests that attention-based and image-language representations are highly competitive in average recognition, but the differences are small enough that they should not be overinterpreted without additional uncertainty analysis.

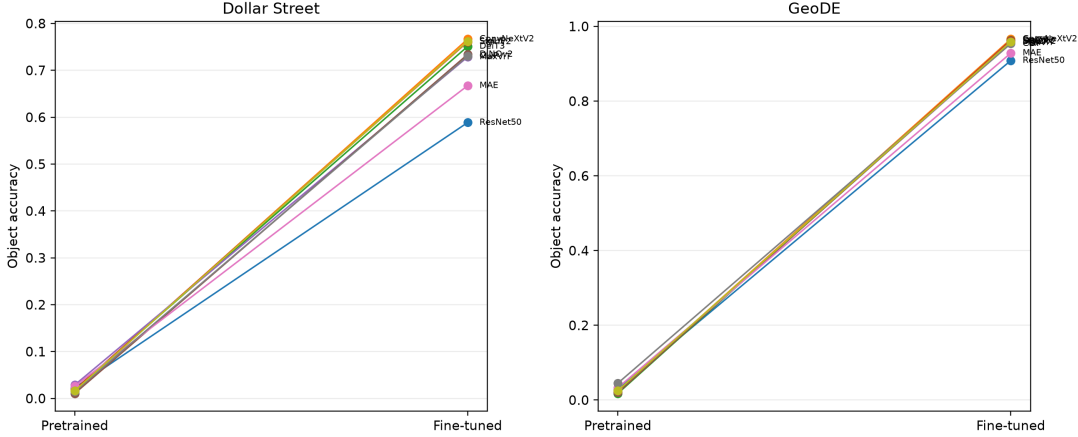


Figure 2: Pretrained-to-fine-tuned change in object accuracy. All evaluated model-dataset pairs move upward after fine-tuning, but the magnitude differs by dataset and model.

Table 4: Paradigm-level summary for fine-tuned runs. Values are percentages.

| Paradigm                            | Avg Acc | Avg Disparity | Best model | Worst model | Main observation                                                                  |
|-------------------------------------|---------|---------------|------------|-------------|-----------------------------------------------------------------------------------|
| Convolutional Vision Paradigm       | 80.8    | 6.4           | ConvNeXtV2 | ResNet50    | ConvNeXtV2 leads accuracy; ResNet50 has the largest mean disparity in this group. |
| Transformer Vision Paradigm         | 85.4    | 5.5           | SwinV2     | MaxViT      | Strong fine-tuned accuracy with mid-range disparity across datasets.              |
| Self-supervised Foundation Paradigm | 82.2    | 4.7           | DINOv2     | MAE         | Lowest average disparity, with MAE trading lower accuracy for the smallest gaps.  |
| Vision-Language Foundation Paradigm | 85.1    | 5.8           | SigLIP     | CLIP        | Competitive accuracy, but CLIP has the largest fine-tuned GeoDE disparity.        |

**Regional disparity beyond average accuracy.** The central result is that high average accuracy does not guarantee low geographic disparity. ConvNeXtV2 has the highest fine-tuned accuracy on both datasets, but it does not have the lowest disparity. MAE has the lowest fine-tuned disparity on both datasets: 6.9% on Dollar Street and 1.9% on GeoDE. By contrast, ConvNeXtV2 has disparity of 8.5% on Dollar Street and 2.3% on GeoDE. This pattern suggests that accuracy and regional balance are distinct empirical properties. It is consistent with worst-group evaluation arguments that aggregate performance can obscure weaker-group behavior [Sagawa et al., 2020], but the result here is specific to geographic regions and should be interpreted descriptively.

The dataset contrast is also clear. Fine-tuned Dollar Street runs have larger geographic disparities than fine-tuned GeoDE runs. The mean fine-tuned disparity is 8.9% on Dollar Street and 2.3% on GeoDE. This may indicate that the processed Dollar Street subset exposes stronger region-conditioned variation for the evaluated models. A safer interpretation is that the two datasets measure different forms of geographic visual variation, and that regional robustness should not be inferred from only one dataset.

**Pretrained versus fine-tuned behavior.** The fine-tuning comparison separates four possible outcomes. The best outcome would be accuracy improvement with disparity reduction. The second outcome is accuracy improvement with disparity increase. The third is accuracy decrease with disparity reduction. The worst outcome is accuracy decrease with disparity increase. In the



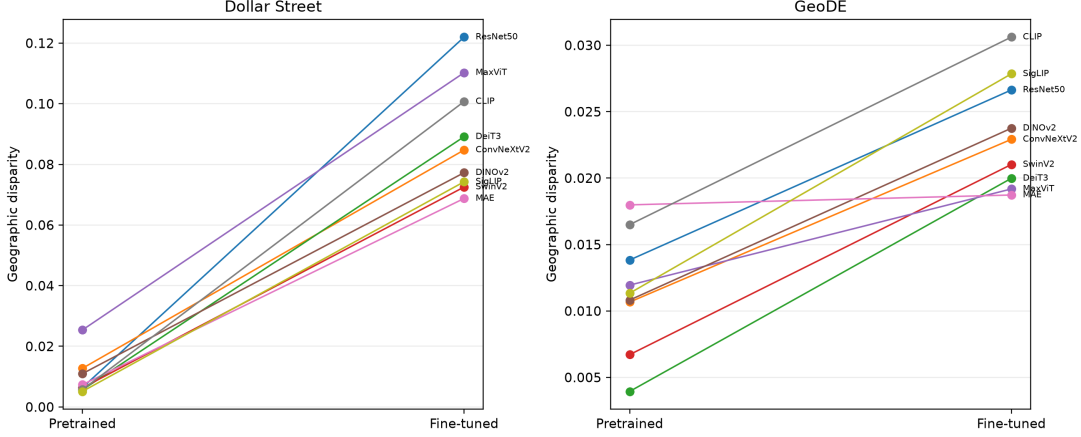


Figure 3: Pretrained-to-fine-tuned change in geographic disparity. All evaluated model-dataset pairs show a larger best-minus-worst regional gap after fine-tuning.

Table 5: Fine-tuned per-region accuracy by model. Values are percentages; – indicates that the region is not used by that dataset.

| Dataset       | Model      | Paradigm                            | Africa | Americas | Asia | EastAsia | Europe | SouthEastAsia | WestAsia | Best region | Worst region  | Gap  |
|---------------|------------|-------------------------------------|--------|----------|------|----------|--------|---------------|----------|-------------|---------------|------|
| Dollar Street | CLIP       | Vision-Language Foundation Paradigm | 67.8   | 74.2     | 73.2 | –        | 77.8   | –             | –        | Europe      | Africa        | 10.1 |
| Dollar Street | ConvNeXtV2 | Convolutional Vision Paradigm       | 72.1   | 78.4     | 76.5 | –        | 80.6   | –             | –        | Europe      | Africa        | 8.5  |
| Dollar Street | DeiT3      | Transformer Vision Paradigm         | 69.7   | 77.5     | 75.2 | –        | 78.6   | –             | –        | Europe      | Africa        | 8.9  |
| Dollar Street | DINOv2     | Self-supervised Foundation Paradigm | 68.9   | 75.9     | 73.1 | –        | 76.7   | –             | –        | Europe      | Africa        | 7.7  |
| Dollar Street | MAE        | Self-supervised Foundation Paradigm | 62.2   | 67.6     | 67.5 | –        | 69.1   | –             | –        | Europe      | Africa        | 6.9  |
| Dollar Street | MaxViT     | Transformer Vision Paradigm         | 66.2   | 74.5     | 73.5 | –        | 77.3   | –             | –        | Europe      | Africa        | 11.0 |
| Dollar Street | ResNet50   | Convolutional Vision Paradigm       | 51.4   | 61.6     | 59.3 | –        | 63.6   | –             | –        | Europe      | Africa        | 12.2 |
| Dollar Street | SigLIP     | Vision-Language Foundation Paradigm | 72.0   | 78.7     | 75.8 | –        | 79.4   | –             | –        | Europe      | Africa        | 7.4  |
| Dollar Street | SwinV2     | Transformer Vision Paradigm         | 71.9   | 78.2     | 75.9 | –        | 79.2   | –             | –        | Europe      | Africa        | 7.3  |
| GeoDE         | CLIP       | Vision-Language Foundation Paradigm | 94.9   | 95.6     | –    | 95.3     | 96.4   | 93.8          | 96.8     | WestAsia    | SouthEastAsia | 3.1  |
| GeoDE         | ConvNeXtV2 | Convolutional Vision Paradigm       | 96.0   | 96.5     | –    | 96.5     | 97.6   | 95.6          | 97.9     | WestAsia    | SouthEastAsia | 2.3  |
| GeoDE         | DeiT3      | Transformer Vision Paradigm         | 96.0   | 96.2     | –    | 96.3     | 97.0   | 95.2          | 97.2     | WestAsia    | SouthEastAsia | 2.0  |
| GeoDE         | DINOv2     | Self-supervised Foundation Paradigm | 95.4   | 96.4     | –    | 96.0     | 96.5   | 94.8          | 97.1     | WestAsia    | SouthEastAsia | 2.4  |
| GeoDE         | MAE        | Self-supervised Foundation Paradigm | 92.2   | 93.2     | –    | 92.6     | 93.6   | 92.0          | 93.9     | WestAsia    | SouthEastAsia | 1.9  |
| GeoDE         | MaxViT     | Transformer Vision Paradigm         | 94.8   | 96.2     | –    | 95.6     | 96.5   | 94.7          | 96.6     | WestAsia    | SouthEastAsia | 1.9  |
| GeoDE         | ResNet50   | Convolutional Vision Paradigm       | 90.1   | 90.9     | –    | 90.3     | 92.4   | 89.7          | 92.2     | Europe      | SouthEastAsia | 2.7  |
| GeoDE         | SigLIP     | Vision-Language Foundation Paradigm | 95.5   | 96.1     | –    | 95.9     | 96.7   | 94.5          | 97.3     | WestAsia    | SouthEastAsia | 2.8  |
| GeoDE         | SwinV2     | Transformer Vision Paradigm         | 95.7   | 96.5     | –    | 95.9     | 97.1   | 95.3          | 97.4     | WestAsia    | SouthEastAsia | 2.1  |

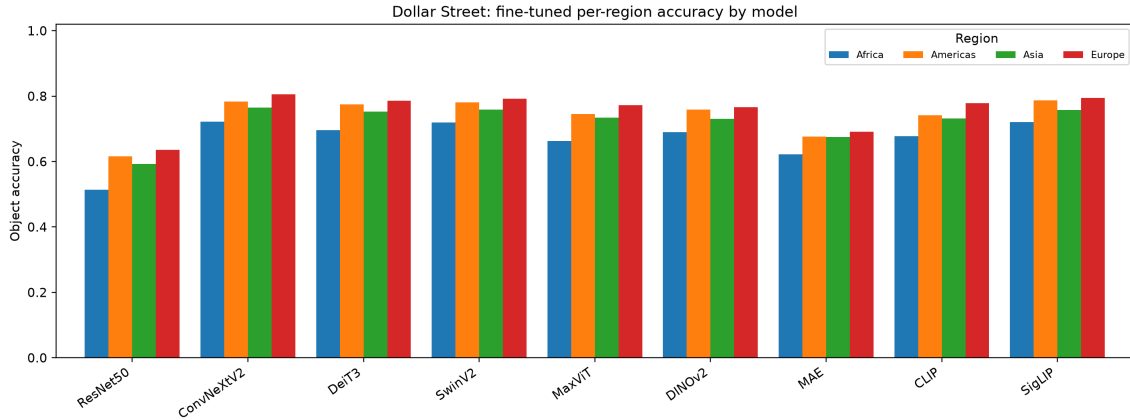


Figure 4: Dollar Street fine-tuned per-region accuracy. Europe is the highest-accuracy region and Africa is the lowest-accuracy region for all nine models in the available result files.

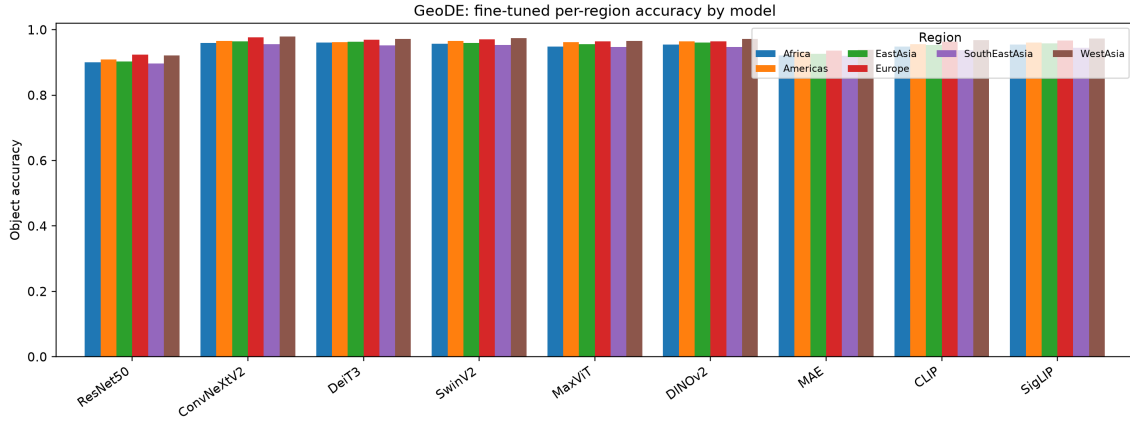


Figure 5: GeoDE fine-tuned per-region accuracy. Most models have WestAsia as the highest-accuracy region and SouthEastAsia as the lowest-accuracy region, although ResNet50 has Europe as its best region.

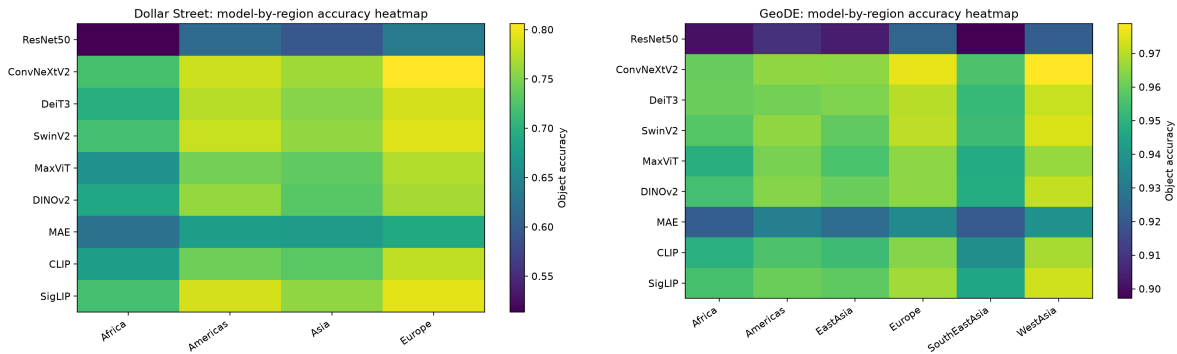


Figure 6: Model-by-region fine-tuned accuracy heatmaps for Dollar Street and GeoDE. The heatmaps make the regional ordering visible without reducing the analysis to a single best-worst pair.

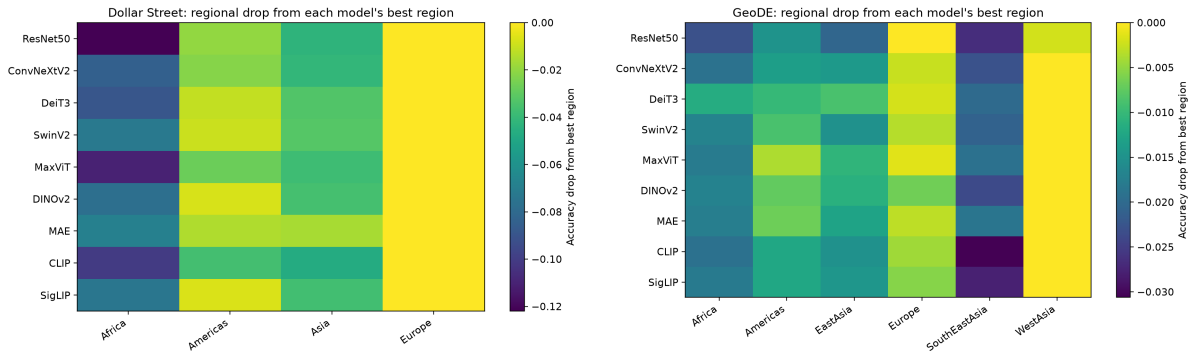


Figure 7: Per-model regional drops relative to each model's best region. This view emphasizes where each model loses accuracy relative to its own strongest region.

current results, every evaluated pair falls into the second category: fine-tuning improves average accuracy and increases measured geographic disparity. This does not prove that fine-tuning causes regional disparity. One possible explanation is that pretrained evaluations are near the floor because of classifier-head and label-mapping constraints, so regional differences are compressed when almost all predictions are wrong. After fine-tuning, the models become useful recognizers, and regional differences become measurable.

The magnitude of disparity increase is larger on Dollar Street than on GeoDE. For example, ResNet50 on Dollar Street increases from 2.3% pretrained accuracy to 58.9% fine-tuned accuracy, while disparity increases from 0.6% to 12.2%. MAE on GeoDE increases from 3.3% to 92.9% accuracy, while disparity changes only from 1.8% to 1.9%. These examples illustrate why the direction of fine-tuning change should be read jointly with the absolute accuracy level.

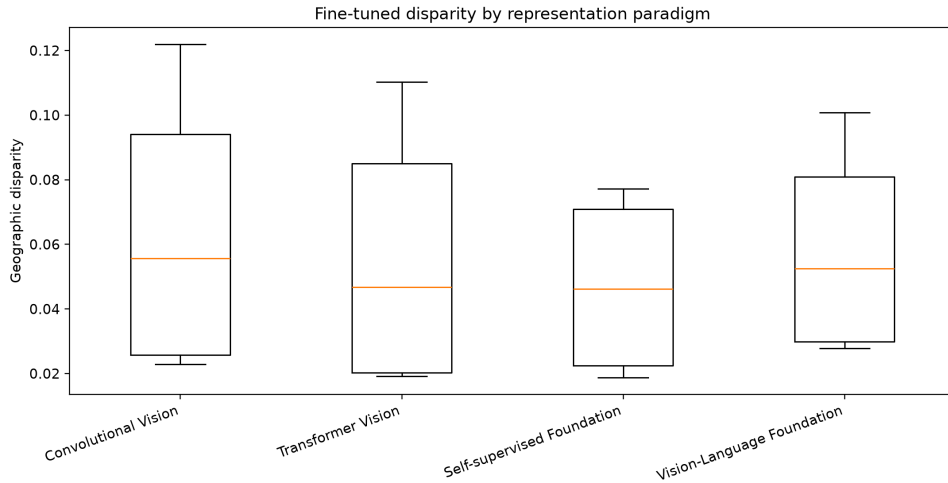


Figure 8: Fine-tuned geographic disparity by representation paradigm. The self-supervised group has the lowest mean disparity in these runs, but the evidence is descriptive and does not establish that self-supervision causes regional robustness.

**Paradigm-level interpretation.** The paradigm-level summary in Table 4 suggests several patterns. The Transformer Vision Paradigm has the highest mean fine-tuned accuracy, while the Self-supervised Foundation Paradigm has the lowest mean fine-tuned disparity. The Vision-Language Foundation Paradigm is competitive in accuracy, but CLIP has the largest fine-tuned GeoDE disparity among all models. The Convolutional Vision Paradigm contains both the strongest average performer, ConvNeXtV2, and the weakest fine-tuned performer, ResNet50, showing that broad family labels can hide important within-family differences.

These patterns may reflect architecture, pretraining objective, data exposure, optimization, or dataset-specific interactions. They should not be read causally. The safer conclusion is that representation paradigms differ in their observed accuracy-disparity profiles, and that regional robustness requires direct measurement rather than inference from model class or average accuracy alone. TODO: add per-class regional gaps and confidence intervals to test whether the same conclusions hold under uncertainty.

## 8 Limitations

This study evaluates only two geographically diverse object-recognition datasets. GeoDE and Dollar Street are highly relevant to the paper’s research question, but they cannot represent all forms of geographic, socioeconomic, cultural, environmental, or imaging variation. The results should therefore be interpreted as evidence from two benchmark settings rather than as a universal statement about regional robustness.

The region labels are coarse. GeoDE uses six world-region groups, while the processed Dollar Street subset uses four region groups. These labels may hide country-level, income-level, cultural, linguistic, environmental, and class-composition differences. A model that appears stable across broad regions could still perform unevenly within a region. Conversely, a broad regional gap could reflect several overlapping factors rather than geography alone.

Fine-tuning can depend on dataset split, sample balance, hyperparameters, and model-selection choices. The current fine-tuning protocol is common across the recorded runs, but the paper does not include larger hyperparameter sweeps, alternative model selection, repeated seeds, or uncertainty intervals. The analysis identifies performance disparities, but it cannot prove the causal source of each disparity.

The ImageNet-compatible mapping simplifies the label space. For Dollar Street, the processed experimental subset uses 64 classes rather than the full official 289-category label space. This mapping may exclude objects, merge concepts, or favor categories that align better with ImageNet-style recognition. The very low pretrained accuracies also mean that pretrained geographic-disparity values should be interpreted cautiously.

The selected nine models represent major paradigms but not all possible architectures or foundation models. ResNet50, ConvNeXtV2, DeiT3, SwinV2, MaxViT, DINOv2, MAE, CLIP, and SigLIP are useful representatives, but each paradigm contains many variants. The paper should not claim that a paradigm as a whole is regionally robust or biased based only on this model set.

Finally, the local experimental metadata still needs one audit before a final version. The Dollar Street SwinV2 fine-tuning configuration contains a stored model-key inconsistency that should be reconciled against the actual run provenance. Until that audit is complete, the SwinV2 results should be treated as part of the intended comparison table but not overused for model-specific claims.

## 9 Conclusion

This paper evaluates regional robustness in object recognition using nine models across four representation paradigms: Convolutional Vision, Transformer Vision, Self-supervised Foundation, and Vision-Language Foundation. The evaluation compares ImageNet-pretrained and fine-tuned states on GeoDE and a processed Dollar Street subset, reporting average accuracy together with per-region accuracy and best-minus-worst geographic disparity.

The results show that fine-tuning substantially improves average accuracy for every evaluated dataset-model pair. The strongest fine-tuned model by average accuracy is ConvNeXtV2, which reaches 76.7% on Dollar Street and 96.6% on GeoDE. However, the same results show that higher average accuracy does not automatically imply lower geographic disparity. MAE has the lowest fine-tuned disparity on both datasets, even though it is not the highest-accuracy model.

Fine-tuning also changes the regional-disparity picture. In all 18 dataset-model pairs, fine-tuning improves average accuracy and increases the measured best-minus-worst regional gap. This finding should be interpreted cautiously because pretrained evaluations are near the accuracy floor,

but it still reinforces the paper’s main message: regional behavior must be measured directly. Dollar Street shows larger fine-tuned regional gaps than GeoDE in these experiments, suggesting that dataset design and visual context can strongly affect conclusions about robustness.

At the paradigm level, the Transformer Vision Paradigm has the highest mean fine-tuned accuracy, while the Self-supervised Foundation Paradigm has the lowest mean fine-tuned disparity. These are descriptive patterns, not causal claims. The broader implication is that benchmark progress and average accuracy alone may not reveal uneven model behavior across geographic contexts. Regional robustness should be evaluated directly, with average accuracy, worst-region accuracy, best-region accuracy, and geographic disparity reported together.

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